

VI. FUGW

Fused Unbalanced Gromov Wasserstein (FUGW) is an approach for aligning pairwise brains based on optimal transformation. We rewrite the Equation (1) of FUGW in [1] as

$$\begin{aligned} \text{FUGW}(\alpha, \beta) := & \theta \cdot L_W(\mathbf{P}) + (1 - \theta) \cdot L_{GW}(\mathbf{P}) + \rho_1 \cdot KL(\mathbf{P}\mathbf{1}|\boldsymbol{\omega}^s) \\ & + \rho_2 \cdot KL(\mathbf{P}^T\mathbf{1}|\boldsymbol{\omega}^t) + \epsilon \cdot E(\mathbf{P}), \end{aligned} \quad (10)$$

where the term $L_W(\mathbf{P})$ maintains the functional information, the $L_{GW}(\mathbf{P})$ preserves the anatomical structure, and θ makes a trade-off between anatomical structure and functional information; moreover, ρ_1, ρ_2 is used to penalize the margin violations, and $E(\mathbf{P})$ is the entropy regularization term. In our experiments, we set $\rho = \rho_1 = \rho_2$.

Its pairwise alignment between α and β is the coupling $\mathbf{P}$ induced by $\text{FUGW}(\alpha, \beta)$. And its corresponding barycenter on $\{\alpha^k\}_{k \in [m]}$ is a measure β that minimizes the following objective

$$\sum_{k=1}^m \text{FUGW}(\alpha^k, \beta). \quad (11)$$

VII. FULL EXPERIMENTS

This section demonstrates the effectiveness of our method for computing pairwise alignments and barycenter. All experiments are performed on a server equipped with 2.40GHz Intel CPU, NVIDIA GeForce RTX 3090 GPU, 64GB main memory, and Python 3.12. Our RWD is executed on the CPU, while FUGW runs on the GPU. Our implementation utilizes some Python libraries [22]–[24].

IBC Dataset: The Individual Brain Charting (IBC)⁵ [25], [26] is a high spatial-resolution, multi-task fMRI dataset. We select 13 subjects from IBC, and 433 fMRI maps for each subject;

ADHD Dataset: Attention Deficit Hyperactivity Disorder (ADHD) dataset is a widely used multi-site neuroimaging dataset. Due to its multi-site nature, the dataset exhibits substantial variability. Thus, it is commonly used to evaluate the robustness of brain analysis. s

We call the above data an original dataset $\mathcal{Q}$. To generate a noisy dataset $\mathcal{Q}'$, we add $\zeta = \zeta_\alpha = \zeta_\beta$ mass of artifacts and apply a random rotation to each original image. The proper margin violation is at most 2ζ .

A. Pairwise alignments

Evaluation metrics: (i) $\text{diff}_a := \|\mathbf{P}\mathbf{1} - \mathbf{a}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{a}$. (ii) $\text{diff}_b := \|\mathbf{P}^T\mathbf{1} - \mathbf{b}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{b}$. (iii) loss_{align} refers to the average distance between pairwise noisy data.

To demonstrate the performance of our pairwise alignment method, we applied our algorithms to the noisy dataset $\mathcal{Q}'$. Each case involves 13 subjects. Thus, we compute $\frac{13 \times 12}{2} = 78$ pairwise alignments between the subjects. The recorded values are the average results for these 78 pairwise alignments. The hyperparameter ρ of FUGW⁶ controls the robustness to noise.

Table III shows that our RWD has better alignment quality than FUGW on noisy datasets, and it is robust to rigid transformations. Moreover, our method does not cause excessive margin violation (i.e., $\text{diff}_a = \text{diff}_b \leq 2\zeta$).

⁵Its original [25] and extended version [26] includes 12 and 15 subjects, respectively.

⁶Code and additional details for FUGW are available on GitHub: iccasp2025.

TABLE III: Comparisons of different alignment methods on noisy IBC dataset. WD is a special case of RWD, where rigid transformation is replaced with identity transformation.

ζ	method	ρ	time(s)	diff _a	diff _b	loss _{align} (↓)
0.1	FUGW	0.02	1626	0.40	0.41	156.70 $\pm$ 61.85
	FUGW	0.03	1626	0.57	0.55	150.55 $\pm$ 59.62
	FUGW	0.06	1626	0.59	0.58	147.28 $\pm$ 59.99
	FUGW	0.13	1616	0.50	0.47	150.64 $\pm$ 59.71
	FUGW	0.25	1623	0.37	0.37	151.63 $\pm$ 61.80
	FUGW	0.50	1627	0.28	0.27	153.14 $\pm$ 63.16
	FUGW	1	1626	0.21	0.21	154.91 $\pm$ 63.65
	FUGW	2	1626	0.16	0.16	157.64 $\pm$ 62.84
	WD	None	328	0.20	0.20	3.35 $\pm$ 1.12
0.2	RWD	None	1475	0.20	0.20	2.42 $\pm$ 0.50
	FUGW	0.02	2029	0.52	0.47	172.96 $\pm$ 67.21
	FUGW	0.03	2023	0.60	0.60	167.30 $\pm$ 67.81
	FUGW	0.06	2021	0.56	0.56	162.62 $\pm$ 68.59
	FUGW	0.13	2020	0.44	0.45	157.83 $\pm$ 64.94
	FUGW	0.25	2021	0.33	0.34	154.73 $\pm$ 67.28
	FUGW	0.50	2023	0.26	0.27	157.49 $\pm$ 66.87
	FUGW	1	2021	0.20	0.21	160.16 $\pm$ 66.20
	FUGW	2	2020	0.14	0.16	159.28 $\pm$ 66.72
0.3	WD	None	398	0.40	0.40	2.72 $\pm$ 1.05
	RWD	None	1746	0.40	0.40	2.01 $\pm$ 0.54
	FUGW	0.02	2650	0.52	0.52	208.50 $\pm$ 92.33
	FUGW	0.03	2651	0.62	0.64	204.80 $\pm$ 94.36
	FUGW	0.06	2651	0.58	0.59	196.32 $\pm$ 93.91
	FUGW	0.13	2652	0.46	0.48	186.11 $\pm$ 92.22
	FUGW	0.25	2652	0.33	0.35	177.40 $\pm$ 91.42
	FUGW	0.50	2653	0.24	0.27	173.17 $\pm$ 91.25
	FUGW	1	2651	0.18	0.21	172.34 $\pm$ 91.35
0.3	FUGW	2	2644	0.13	0.17	173.02 $\pm$ 91.79
	WD	None	487	0.59	0.59	3.84 $\pm$ 2.39
	RWD	None	2152	0.59	0.59	1.74 $\pm$ 0.73

Figure 2 presents the visualization of different alignment methods with $\zeta = 0.1$ on IBC dataset. It can be observed that our RWD-based alignment method achieves better performance. This advantage is particularly evident in the rightmost column.

Specifically, WD-based methods fails to effectively suppress noise, indicating limited robustness to noise. The FUGW-based approach needs search the hyperparameter ρ , which is difficult to tune in practice for varying mass of noise; moreover, the diffusion effect introduced by this method leads to noticeable information degradation.

In contrast, our RWD-based alignment method achieves more accurate alignments while preserving more details, demonstrating superior robustness and stability under noisy conditions.

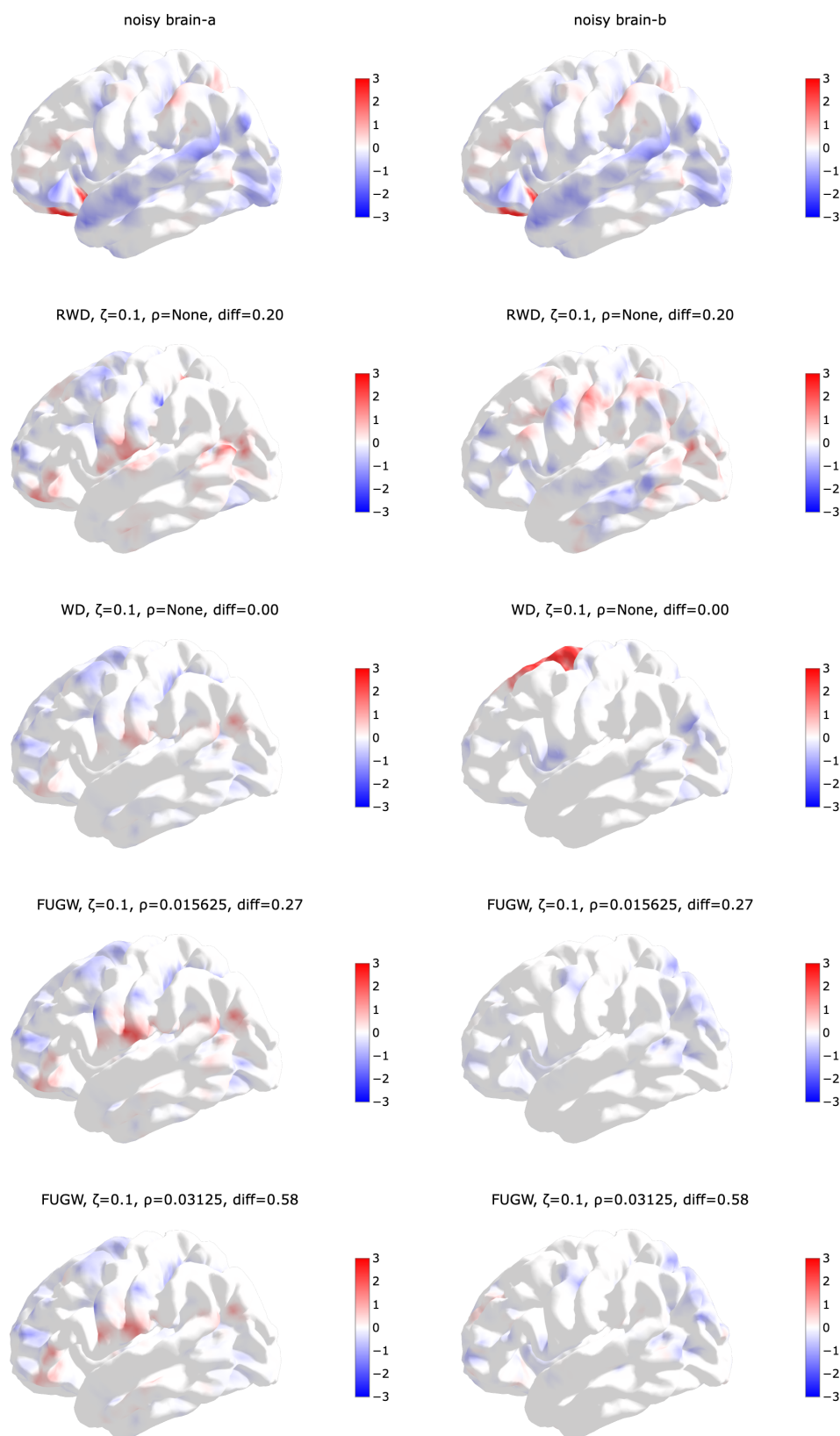


Fig. 2: Visualization of alignment methods with $\zeta = 0.1$ on IBC dataset.

TABLE IV: Comparisons of different barycenter algorithms on noisy IBC dataset.

ζ	method	ρ	time(s)	diff	loss($\downarrow$)
0.1	FUGW	4	571	1.11	66.70 $\pm$ 26.62
	FUGW	8	571	0.64	67.28 $\pm$ 26.66
	FUGW	12	573	0.42	67.09 $\pm$ 26.64
	FUGW	16	572	0.31	67.52 $\pm$ 26.79
	FUGW	20	569	0.24	67.22 $\pm$ 26.85
	FUGW	24	569	0.20	67.07 $\pm$ 27.07
	FUGW	28	569	0.17	67.46 $\pm$ 26.84
	FUGW	32	572	0.15	67.66 $\pm$ 26.84
	FUGW	36	572	0.14	67.43 $\pm$ 27.01
	FUGW	40	572	0.12	67.44 $\pm$ 26.87
	WD	None	125	0.20	68.37 $\pm$ 28.54
	RWD	None	467	0.20	8.43 $\pm$ 0.17
0.2	FUGW	4	640	1.18	24.74 $\pm$ 5.91
	FUGW	8	640	0.73	24.72 $\pm$ 5.98
	FUGW	12	640	0.48	23.80 $\pm$ 6.01
	FUGW	16	640	0.35	23.58 $\pm$ 5.72
	FUGW	20	642	0.27	23.45 $\pm$ 5.84
	FUGW	24	641	0.22	23.61 $\pm$ 5.94
	FUGW	28	641	0.19	23.55 $\pm$ 5.79
	FUGW	32	642	0.16	23.59 $\pm$ 6.03
	FUGW	36	641	0.15	23.71 $\pm$ 5.90
	FUGW	40	640	0.13	23.90 $\pm$ 5.95
	WD	None	139	0.40	19.20 $\pm$ 5.52
	RWD	None	502	0.40	1.10 $\pm$ 0.11
0.3	FUGW	4	731	1.12	160.09 $\pm$ 43.12
	FUGW	8	732	0.82	132.55 $\pm$ 39.10
	FUGW	12	732	0.56	117.68 $\pm$ 36.00
	FUGW	16	732	0.40	111.21 $\pm$ 34.96
	FUGW	20	732	0.31	108.37 $\pm$ 34.58
	FUGW	24	731	0.25	107.41 $\pm$ 34.69
	FUGW	28	732	0.21	106.96 $\pm$ 34.41
	FUGW	32	730	0.18	106.39 $\pm$ 34.50
	FUGW	36	731	0.16	105.28 $\pm$ 34.50
	FUGW	40	732	0.14	105.26 $\pm$ 34.40
	WD	None	148	0.60	94.73 $\pm$ 32.63
	RWD	None	562	0.60	1.19 $\pm$ 0.12

B. Barycenter

Evaluation metrics: (i) *diff* records the average deviation of couplings from the weights of m input data and the barycenter. (ii) *loss* represents the cost incurred from the barycenter to the clean data.

Table IV and Table V show the results of our barycenter algorithm on IBC dataset and ADHD dataset. We select six data from the noisy dataset $\mathcal{Q}'$ as input. We compare our barycenter method (RWD-BC) with its corresponding version without rigid transformation (WD-BC) and the FUGW-based method (FUGW-BC). Table II shows that our RWD-BC yields a higher quality barycenter across varying levels of noise.

Figures 3 to 5 illustrate the z-scores of the barycenters on IBC dataset, and Figures 6 to 8 show the results on ADHD dataset. Compared to FUGW-BC and WD-BC, our RWD-BC captures finer details. The FUGW-BC, which incorporates an entropy regularization term, exhibits a diffusion-like behavior when applied to voxel fMRI images (as seen in Figure 1). In FUGW [1], this diffusion effect is mitigated by mapping voxel images onto a pre-specified template before calculating the barycenter, making the method dependent on pre-specified template. In contrast, our method does not include the entropy regularization term, allowing it to directly handle voxel images without the need for pre-specified templates.

In conclusion, our method achieves better alignment quality and produces a finer barycenter without relying on any pre-specified template. Moreover, it eliminates the need for tuning the parameter ρ to handle varying levels of noise and does not result in excessive marginal violations. Additionally, our algorithm is more computationally efficient on a CPU than FUGW on a GPU, saving both time and hardware resources.

TABLE V: Comparisons of different barycenter algorithms on noisy ADHD dataset.

ζ	method	ρ	time(s)	diff	$loss(\downarrow)$
0.2	FUGW	4	1938	0.85	4201590 ± 229009
	FUGW	12	1724	0.47	6629606 ± 216254
	FUGW	16	1602	0.32	7785562 ± 209044
	FUGW	16	1739	0.25	8442489 ± 205096
	FUGW	20	1559	0.20	8835359 ± 200088
	FUGW	24	1671	0.17	9131148 ± 200301
	FUGW	28	1717	0.14	9329023 ± 197472
	FUGW	32	1736	0.12	9502731 ± 195819
	FUGW	36	1636	0.11	9619077 ± 197257
	FUGW	40	1598	0.10	9731046 ± 193806
	WD	None	74	0.38	7754 ± 289
	RWD	None	189	0.40	6223 ± 355
0.3	FUGW	4	1583	0.85	4208795 ± 228898
	FUGW	8	1386	0.48	6629953 ± 215310
	FUGW	12	1239	0.32	7787488 ± 208281
	FUGW	16	1381	0.25	8444297 ± 205368
	FUGW	20	1555	0.20	8841865 ± 201912
	FUGW	24	1268	0.17	9129032 ± 199587
	FUGW	28	1276	0.14	9330253 ± 198285
	FUGW	32	1243	0.12	9486310 ± 197715
	FUGW	36	1237	0.11	9615322 ± 197571
	FUGW	40	1331	0.10	9728489 ± 195238
	WD	None	69	0.57	12471 ± 344
	RWD	None	401	0.60	5505 ± 133
0.4	FUGW	4	1841	0.85	4208651 ± 29028
	FUGW	8	1580	0.48	6629806 ± 15218
	FUGW	12	1544	0.32	7787339 ± 08495
	FUGW	16	1403	0.25	8444145 ± 05491
	FUGW	20	1417	0.20	8841713 ± 01806
	FUGW	24	1386	0.17	9128879 ± 99725
	FUGW	28	1300	0.14	9330099 ± 98307
	FUGW	32	1349	0.12	9486156 ± 97716
	FUGW	36	1318	0.11	9615168 ± 97515
	FUGW	40	1303	0.10	9728335 ± 95260
	WD	None	62	0.77	49221 ± 3441
	RWD	None	360	0.79	4648 ± 235

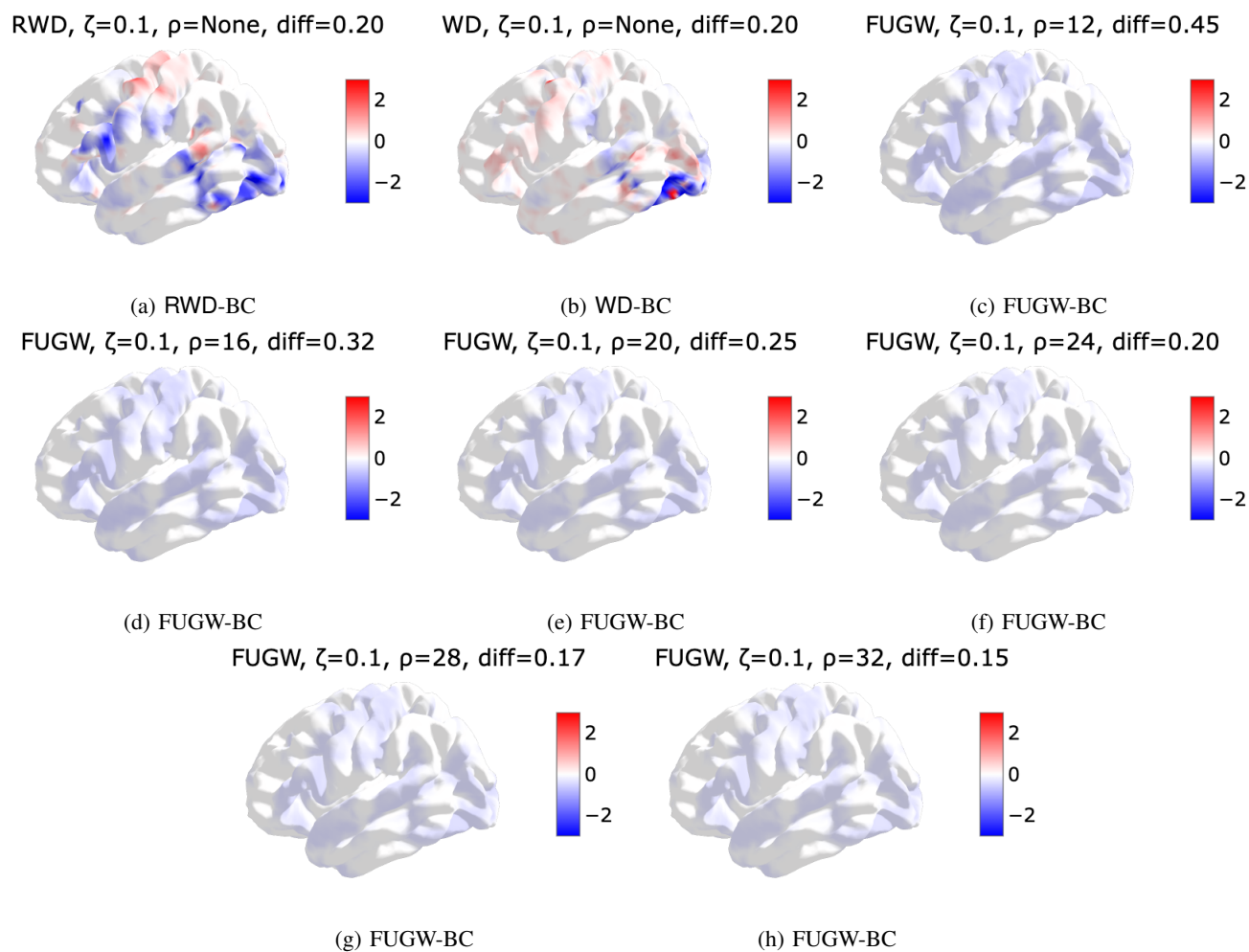
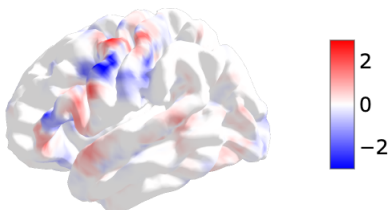


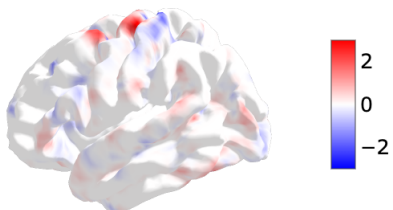
Fig. 3: z-score of barycenters with $\zeta = 0.1$ on IBC dataset.

RWD, $\zeta=0.2$, $\rho=\text{None}$, $\text{diff}=0.40$



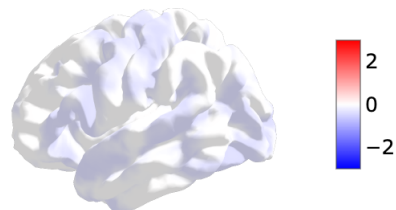
(a) RWD-BC

WD, $\zeta=0.2$, $\rho=\text{None}$, $\text{diff}=0.40$



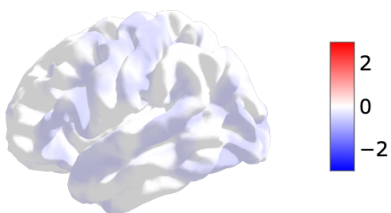
(b) WD-BC

FUGW, $\zeta=0.2$, $\rho=12$, $\text{diff}=0.48$



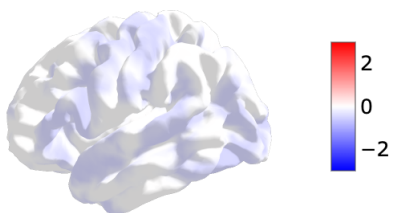
(c) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=16$, $\text{diff}=0.34$



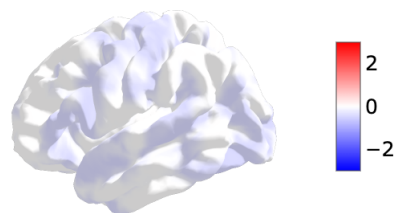
(d) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=20$, $\text{diff}=0.27$



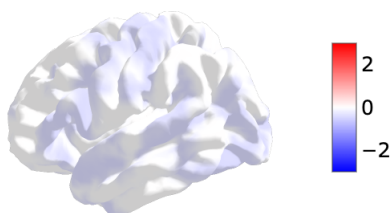
(e) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=24$, $\text{diff}=0.22$



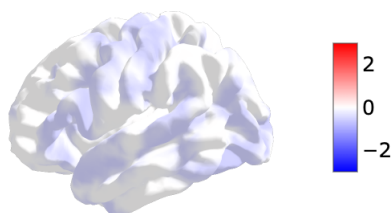
(f) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=28$, $\text{diff}=0.19$



(g) FUGW-BC

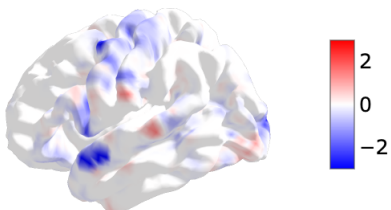
FUGW, $\zeta=0.2$, $\rho=32$, $\text{diff}=0.16$



(h) FUGW-BC

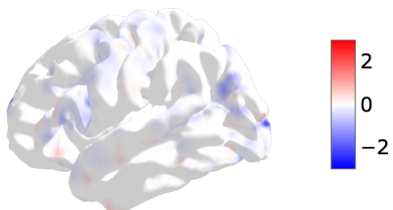
Fig. 4: z-score of barycenters $\zeta = 0.2$ on IBC dataset.

RWD, $\zeta=0.3$, $\rho=\text{None}$, $\text{diff}=0.60$



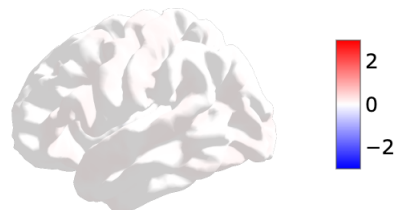
(a) RWD-BC

WD, $\zeta=0.3$, $\rho=\text{None}$, $\text{diff}=0.60$



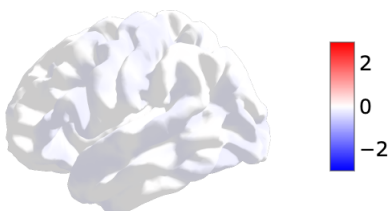
(b) WD-BC

FUGW, $\zeta=0.3$, $\rho=12$, $\text{diff}=0.53$



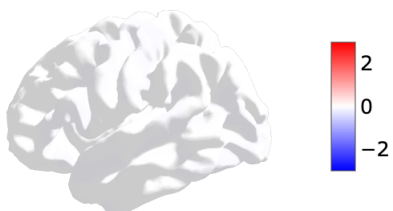
(c) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=16$, $\text{diff}=0.39$



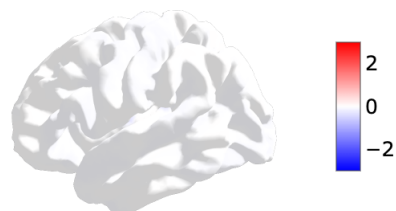
(d) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=20$, $\text{diff}=0.30$



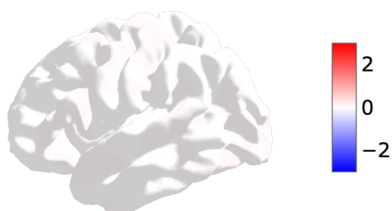
(e) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=24$, $\text{diff}=0.24$



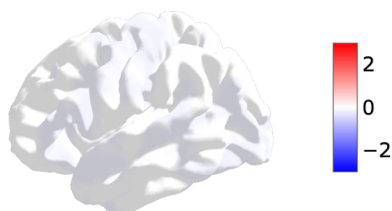
(f) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=28$, $\text{diff}=0.21$



(g) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=32$, $\text{diff}=0.18$



(h) FUGW-BC

Fig. 5: z-score of barycenters $\zeta = 0.3$ on IBC dataset.

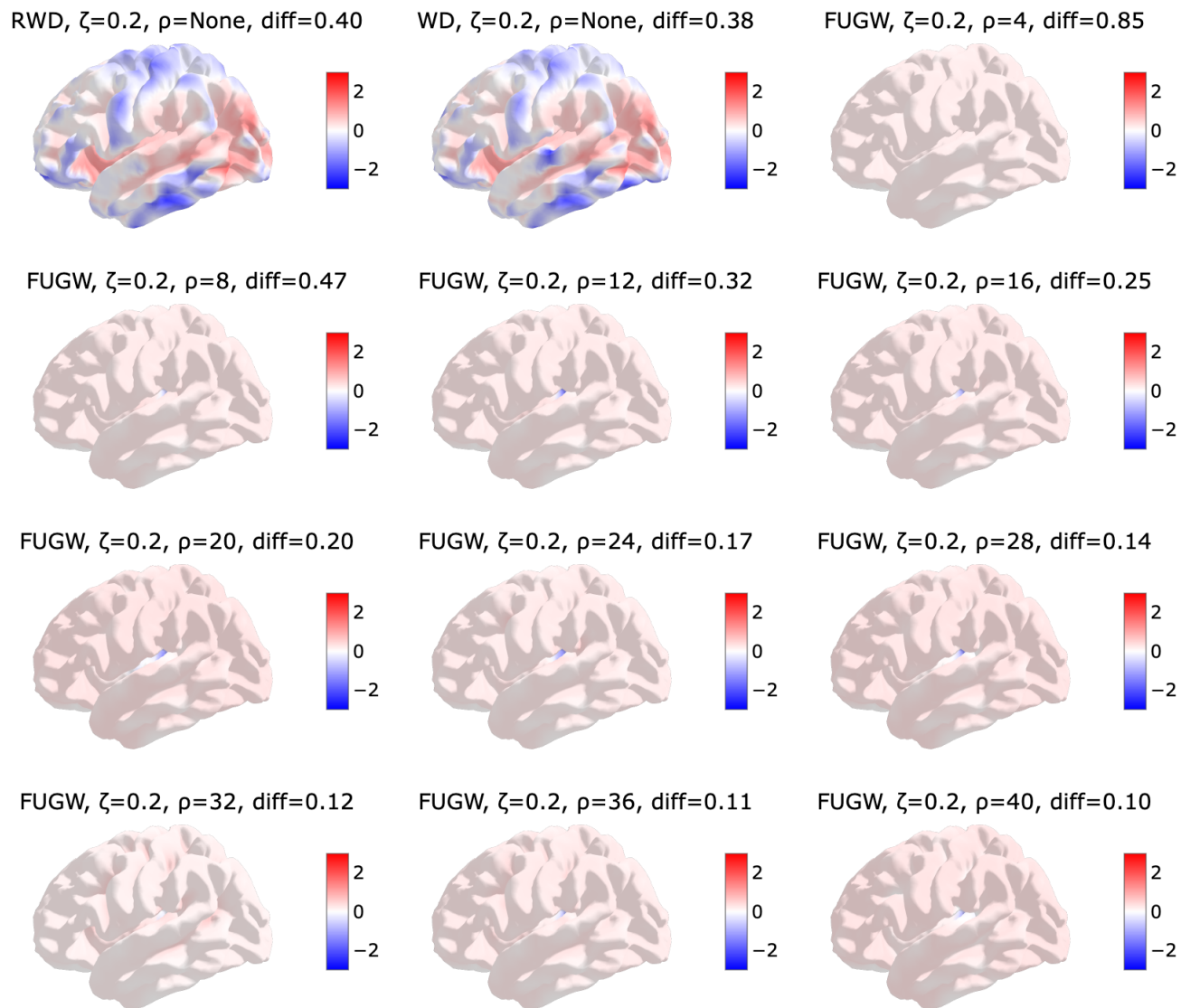


Fig. 6: z-score of barycenters $\zeta = 0.2$ on ADHD dataset.

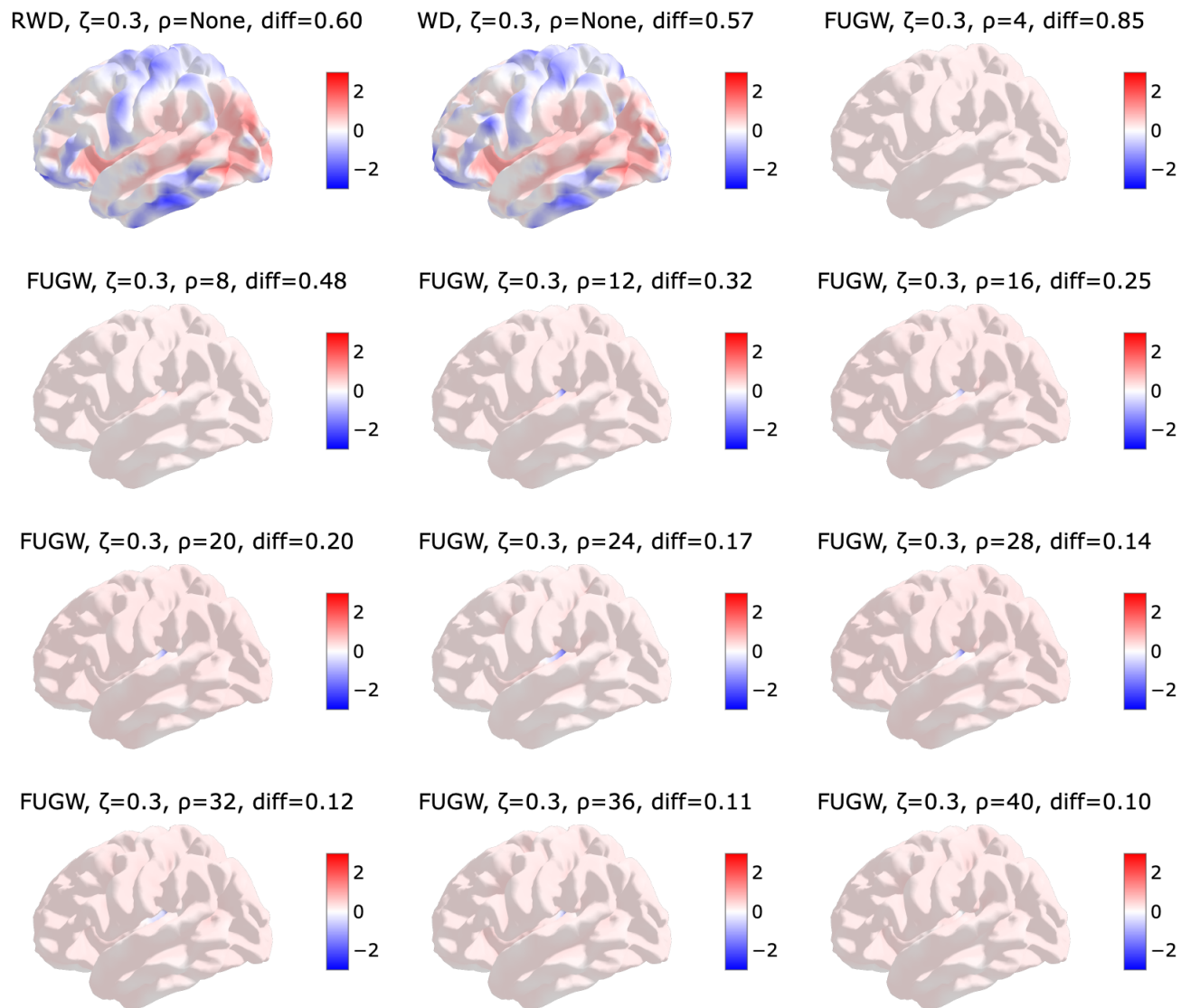


Fig. 7: z-score of barycenters $\zeta = 0.3$ on ADHD dataset.

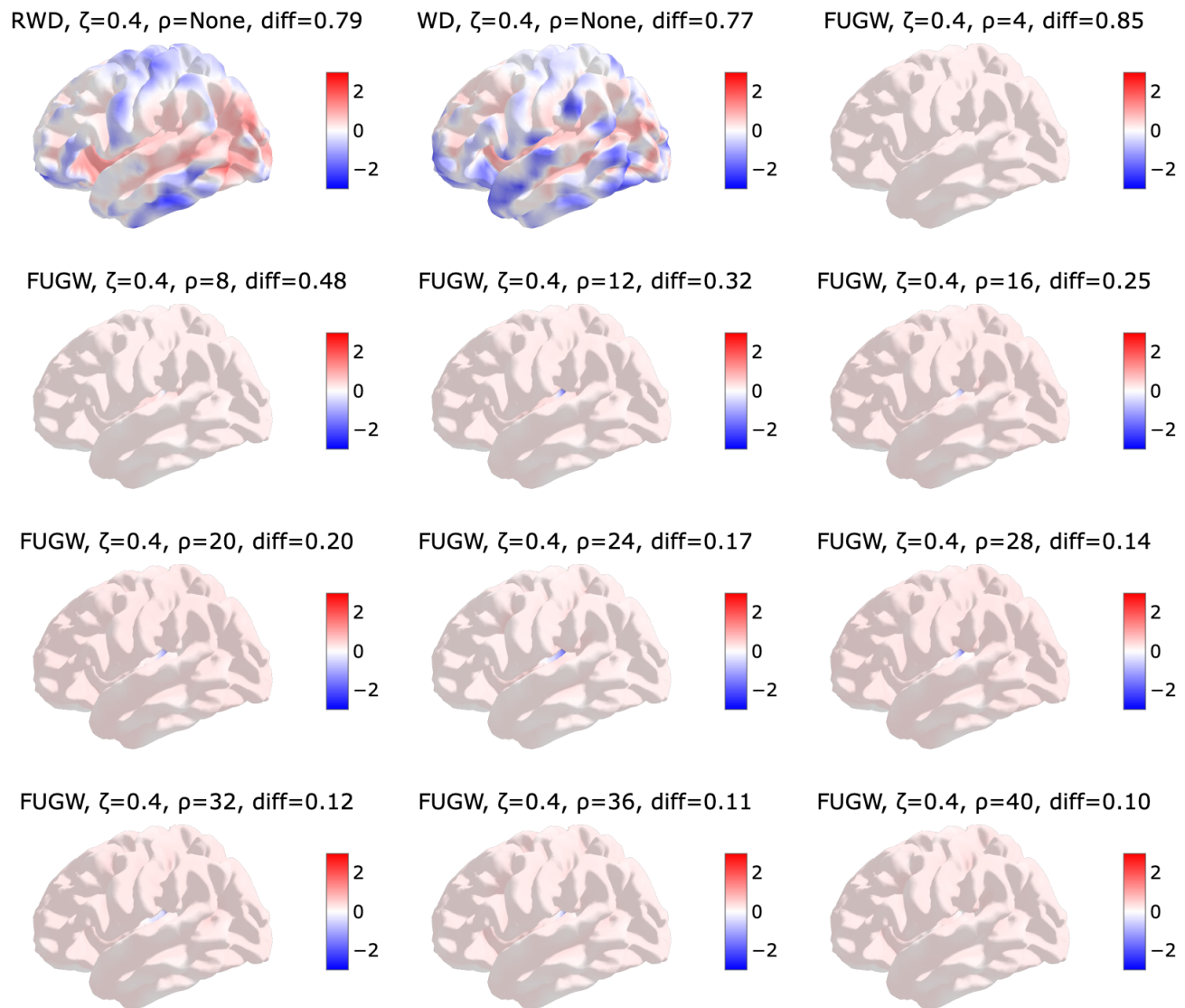


Fig. 8: z-score of barycenters $\zeta = 0.4$ on ADHD dataset.